TIME SERIES AND RANDOM PROCESSES IN LINEAR SYSTEMS MTH3230
MTH3230 Assignment 3

Due date: Week 12, Sunday 25 October 2026

Project Overview
Build, evaluate, and explain a deep learning model to forecast a univariate time series.
Students must:

- Frame a clear forecasting task (horizon, granularity, stakeholders, success criteria).
- Construct supervised learning datasets from time series (sliding windows).
- Train, tune, and diagnose a deep learning forecaster.
- Communicate results and limitations for a non-technical audience.

Collaboration
You can work independently or in a team of up to 3 people. For teamwork, each member should submit the same report, with the names of all team members included for record.

Submission
Python code and a short report (PDF) including:

- Problem statement and dataset description, including the target variable, sampling frequency, preprocessing, and train/validation/test split
- Model design summary (architecture, hyperparameters, optimizer, activation functions, and regularization)
- Performance results (table of train/validation/test errors, comparison with a simple baseline, test-set forecast plot, and learning curves)
- Interpretation, limitations, and possible next steps

Method:
You may use a Multi-Layer Perceptron (MLP), one-dimensional Convolutional Neural Network (CNN), or Recurrent Neural Network (RNN), including LSTM or GRU networks, to implement your Python forecasting model.
Forecasting protocol (applies to all model choices):

- Define the forecasting horizon and input window clearly.
- Split the data chronologically into training, validation, and test sets. Do not randomly split observations across time.
- Fit scaling/normalization and any other data-dependent preprocessing using the training set only, then apply the fitted transformation to validation and test data.
- Construct sliding-window samples without using future observations in the inputs.
- Compare the deep learning model with at least one simple forecasting baseline (for example, a persistence/naive forecast or a seasonal naive forecast when appropriate).
- Report errors on the original scale of the target after reversing any normalization. Use MAE and MSE or RMSE; MASE is strongly encouraged because it is scale-free.

If you choose MLP: You are encouraged to investigate how the following design choices influence forecasting accuracy:

- Input window size (length of historical data used as model input)
- Amount of training history used (for example, recent history vs. the full available training period)
- Network architecture: number of hidden layers and units per layer

- Activation functions (e.g., ReLU, tanh, GELU)
- Optimizer and its parameter settings (e.g., Adam, SGD, RMSprop)
- Regularization techniques (for example, Dropout and L1/L2 weight penalties)
- Evaluation metrics (for example, MAE, MSE/RMSE, and MASE)
- Batch size and learning rate
- Early stopping strategy and patience parameter

If you choose CNN: Use a one-dimensional CNN (Conv1D) and explore how the following design choices affect model performance:

- Input window size (sequence length)
- Amount of training history used (recent history vs. the full available training period)
- Convolution kernel size and, if relevant, stride/dilation
- Number of filters / channels
- Convolutional depth (number of stacked Conv1D layers)
- Pooling strategy (MaxPooling, AveragePooling, or none)
- Activation functions
- Learning rate and optimizer choice
- Regularization (Dropout, L2)
- Flatten vs. GlobalPooling before the final dense regression output
- Batch size and learning rate
- Early stopping strategy and patience parameter

If you choose RNN: You may use a Simple RNN, LSTM, or GRU. You are encouraged to investigate how the following design choices affect model performance:

- Input window size (sequence length)
- RNN cell type (Simple RNN, LSTM, or GRU)
- Number of recurrent layers and hidden units per layer
- Sequence-to-one output design for one-step forecasting; for stacked recurrent layers, specify how intermediate sequences are passed between layers
- Dropout and, where supported, recurrent dropout
- Activation functions and recurrent activation settings, where applicable
- Batch size and learning rate
- Optimizer choice and parameter settings
- Early stopping strategy and patience parameter
- Gradient clipping, if training becomes unstable or gradients explode
- Stateful vs. stateless training, if investigated; clearly explain how hidden states are reset between independent sequences

Optional Advanced Experiments (Bonus Credit)

1. Model comparison: Evaluate and compare two or more architectures (MLP, CNN, RNN/LSTM/GRU) on the same dataset.
2. Hybrid architecture: Combine architectures, for example CNN for local feature extraction followed by an RNN/LSTM/GRU or MLP for forecasting.
3. Multi-step forecasting: Predict multiple future time steps simultaneously (recursive, direct, or multi-output).
4. Feature augmentation: Add exogenous variables such as calendar, weather, or external indicators to improve accuracy.

Note: You are not required to test all variations listed above. You may select the most relevant ones for your dataset, or explore additional variations beyond this list if you wish.

Dataset Options (choose one)
For the core assignment, the output must be a single continuous time-series target. Other variables may be used only if you choose the optional feature-augmentation experiment.

1. Household Electricity Consumption
Measurements of active/reactive power, voltage, and energy sub-metering from a single household (2006–2010).
Forecasting task: Select one continuous target series (for example, global active power or one sub-metering load) and forecast its future value.
https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

2. Retail Sales (US Census)
Official US Census monthly retail sales data by industry group, seasonally adjusted.
Covers 1992–present.
Forecasting task: Predict future retail sales for a sector.
https://www.census.gov/retail/marts/www/timeseries.html

3. Weather – Australian Daily Climate
Daily weather observations from many Australian stations, including temperature, rainfall, humidity, and wind variables.
Forecasting task: Choose one station and one continuous target series (for example, maximum temperature, minimum temperature, or rainfall amount) and forecast its next value. Do not use the RainTomorrow classification target for this assignment.
https://www.kaggle.com/datasets/jsphyg/weather-dataset-rattle-package

4. Energy Consumption – PJM Hourly Electricity Load
Hourly load (MW) for the PJM interconnection grid in the eastern US, 2002–2018.
Forecasting task: Forecast hourly or daily electricity demand.
https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption

5. Stock Market – Apple Inc. (AAPL) Yahoo Finance API via yfinance
Daily open/high/low/close/volume data for Apple (AAPL), available through yfinance.
Forecasting task: Forecast the next-day adjusted closing price or return. If forecasting price, compare against the persistence baseline in which tomorrow's price is predicted by today's price.
https://pypi.org/project/yfinance/

6. Commodity Prices – Crude Oil (WTI)
Daily West Texas Intermediate (WTI) crude oil spot price in USD per barrel since 1986.
Forecasting task: Forecast the next available daily price or return.
https://fred.stlouisfed.org/series/DCOILWTICO

7. Economic Indicators – Unemployment Rate (US)
Monthly US unemployment percentage, 1948–present.
Forecasting task: Forecast unemployment rate 1–6 months ahead.
https://fred.stlouisfed.org/series/UNRATE